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## *In Situ* Brain Perfusion Technique

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### 1. Introduction

Many techniques have been developed to study transport across the blood–brain barrier (BBB). One that offers particular advantage is the *in situ* brain perfusion technique. The primary objective of the *in situ* brain perfusion technique is to take over the circulation to the brain via direct infusion of artificial blood, plasma, or saline into the heart or major vessels leading to the brain, such as the carotid arteries ***(1–4)***. The solute of interest is mixed into the perfusion medium at a known concentration and delivered to the brain via the perfusion fluid for a defined interval. At preselected times, the perfusion is stopped and the amount of solute in brain is determined. From these measurements, the kinetics of brain uptake are analyzed and appropriate transport or permeability constants calculated ***(5)***.

There are a number of advantages of this approach for studies of BBB transport. First, the kinetics of brain uptake via perfusion are quite simple because the compound of interest is delivered to the brain at a set, known concentration that does not vary with time. Second, because of the investigator's control of perfusion uptake time, transport can be measured over a broad range of intervals (anywhere from 5 s to >30 min), allowing accurate determination of permeability over a $10^5$ range ***(2)***. Third, because solute is not delivered first to peripheral metabolizing organs (e.g., liver, lung or kidney), the investigator can generally be confident that the compound is presented to brain in the intact form and thus minimizes chances for error due to brain uptake of permeable metabolites, especially $^3H\text{-}H_2O$ and $^{14}C\text{-}CO_2$ ***(6,7)***. Fourth, because the concentration of solute as well as potential competitors, blockers, ions (e.g., $Na^+$), and plasma proteins can be readily manipulated in the perfusion medium, the method is ideal for in vivo characterization of the kinetics of saturable transport across the BBB and the effects of plasma protein binding ***(4,8–10)***. And finally, the method can be used to study solute efflux from brain after preloading by switching the perfusate to solute-free saline ***(3,10)***.

A number of perfusion systems have been published over the past 20 yr for study of drug transport into brain. This chapter will present the most widely used method—the *in situ* rat brain perfusion technique (or "Takasato" method).

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## 2. Materials

### 2.1. Surgical Preparation

1. Sodium pentobarbital, or a suitable animal anesthetic (Sigma Chemical, St. Louis, MO).
2. Surgical shaver (e.g., electric clippers) (Stoelting, Wood Dale, IL or Fisher Scientific, Pittsburgh, PA).
3. Heating blanket or light—to maintain rat body temperature during surgery (Stoelting or Harvard Apparatus, Holliston, MA).
4. Indicating Controller (Yellow Springs Instruments, Yellow Springs, OH).
5. Temperature probe (Harvard Apparatus, Stoelting or Yellow Springs Instruments).

### 2.2. Surgery

1. Rodent surgical board (Harvard Apparatus or Stoelting).
2. Binocular dissecting microscope (up to power 40x) (Carl Zeiss Surgical, Thornwood, NY).
3. Fiber optic light illuminator—for dissecting microscope (Carl Zeiss Surgical).
4. Surgical instruments (e.g., microdissecting tweezers and forceps, retractors, microvascular clamps and applicator, spring scissors, microdissecting scissors, operating scissors, retractors, hemostatic forceps, and scalpel blades and handles) (Roboz Surgical Instruments, Rockville, MD; Daigger, Vernon Hills, IL).
5. Surgical silk (4-0, 6-0) (Ethicon, Johnson & Johnson Health Care Systems, Piscataway, NJ).
6. Polyethylene tubing (PE- 60 or PE-50)—for catheters (Becton Dickinson, Sparks, MD).
7. Plastic syringes (1 mL and 5 mL) (Becton Dickinson).
8. Syringe needles—20-gage for PE-60 tubing and 23-gage for PE-50 (Becton Dickinson).
9. Plastic ultra three-way stopcocks (Medex, Hilliard, OH) or four-way stopcock (Hamilton, Reno, NV).
10. Heparin (Sigma, St. Louis, MO).
11. Bacteriostatic 0.9% sodium chloride solution (Abbott Laboratories, North Chicago, IL).
12. Malis Bipolar coagulator (Codman, Raynham, MA).
13. Cotton-tipped applicators (Allegiance, MacGraw Park, IL).
14. Absorbent paper (Fisher Scientific).

### 2.3. Perfusion Fluid

1. Perfusion buffer or fluid: If physiologic saline fluid is used, excellent results are obtained with a perfusion buffer of the following composition: 128 m*M* NaCl, 24 m*M* $NaHCO_3$, 4.2 m*M* KCl, 2.4 m*M* $NaH_2PO_4$, 1.5 m*M* $CaCl_2$, and 0.9 m*M* $MgCl_2$ (Sigma or Fisher Scientific).
2. D-Glucose: added to perfusion fluid on day or experiment (Sigma).
3. Bovine serum albumin or 70 k dextran if oncotic agent is used; added to perfusion fluid on day or experiment (Sigma).
4. Red cells: collected fresh on the day of the experiment in physiologic blood perfusions.
5. 95% oxygen/5% carbon dioxide gas.
6. Glass infusion syringe; 5–50 mL, depending on required infusion rate for infusion pump (Becton Dickinson).
7. Circulating water bath: capable on holding temperature of perfusion fluid and syringes at 37–39° C (Lauda, Daigger or Fisher Scientific).

### 2.4. Perfusion Procedure

1. Constant or variable rate infusion pump (Harvard Apparatus, Holliston, MA).
2. Physiologic pressure transducer (Gould, Valleyview, OH).
3. Amplifier and chart recorder for pressure transducer (Gould).

4. Microcentrifuge: for centrifuging blood samples (Fisher Scientific).
5. Analytical balance: for weighing samples (Sartorius, Daigger).
6. Kimwipes (Kimberly-Clark, Roswell, GA).
7. Ice bucket with ice (Fisher Scientific).
8. Glass petri dish (12–20 cm): to dissect brain on (Fisher Scientific).
9. Filter paper (Whatman, Fisher Scientific).
10. Vials (plastic or glass): for sample collection and holding (Fisher Scientific).
11. Stop watch or timer (Fisher Scientific or Daigger).

## 3. Methods

The specific steps required to carry out a brain perfusion experiment are described from the point of anesthesia and surgical preparation of the animal through sample collection and data analysis

### *3.1. Surgical Preparation*

1. Anesthetize an adult rat (body weight 220–380 g) with sodium pentobarbital (40–50 mg/kg, 50–60 mg/mL, intraperitoneally [i.p.]) or other suitable anesthetic agent.
2. Once surgical anesthesia is obtained, the neck of the animal is shaved using electric clippers and the area is cleaned in preparation for surgery.
3. Attach the animal to the surgical board taking care not to inhibit respiration or blood flow to the head, limbs or thoracic cavity.
4. The animal's body temperature is maintained at 37°C during surgery and perfusion through the use of a heating pad or lamp that is linked to a feedback device (Indicating Controller) with a temperature probe placed in the rectum.

### *3.2. Surgery*

1. A surgical incision is made on the ventral surface of the neck and the subcutaneous tissue and muscles are moved laterally to expose either the right or left carotid artery system (**Fig. 1,** *see* **Note 1**). Choice or right or left carotid artery depends on preference of the surgeon.
2. The common carotid artery is encircled and ligated with surgical silk (4-0) and then a cannula (15 cm of PE-60 containing 100 IU/mL heparin in 0.9% NaCl) is placed in the common carotid artery (*see* **Fig. 1**) distal to the ligature for delivery of perfusion fluid to brain (*see* **Notes 2** and **3**).
3. The corresponding external carotid artery is also ligated with surgical silk (6-0) to reduce the volume of perfusion fluid that goes to extracerebral tissues.
4. At the end of the surgical procedure, the wound site is cleaned and body rectal temperature is checked to ensure that it is approx 37°C. Rat rectal temperature is maintained following surgery through the use of a heating pad or heat lamp connected to a feedback device and a rectal probe.

### *3.3. Perfusion Fluid*

1. The most frequently used perfusion fluid is a bicarbonate-buffered physiologic saline ***(8)*** (*see* **Subheading 2.3.** and **Note 4**).
2. Five to 10 m*M* D-glucose is usually added to the perfusate on the day of the experiment to maintain cerebral glucose metabolism and adenosine triphosphate (ATP) production.
3. Plasma oncotic agents and oxygen-carrying agents are also added as required (*see* **Note 5**).
4. Care should be taken to filter the perfusion medium (0.2 μm) and heat it to 37°C.
5. Filtered erythrocytes (if used) are added to a hematocrit of 10–40% ***(2,3)***(*see* **Note 6**).

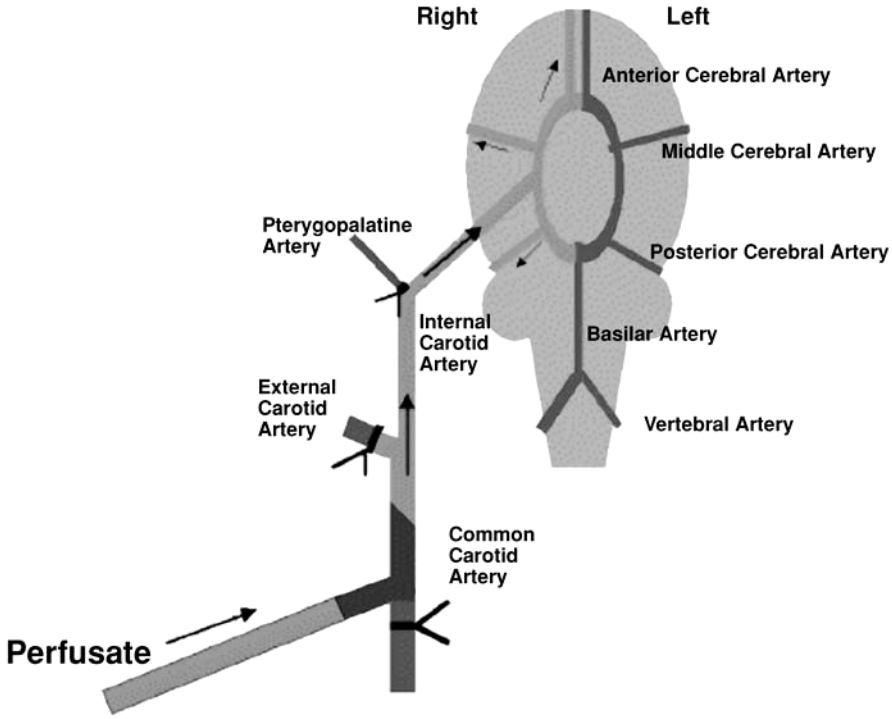


Fig. 1. Diagram of the modified Takasato method with catheter placement in the common carotid artery and ligation of the external carotid artery. The pterygopalatine artery can be either ligated or left open ***(6,10,12)***. The heart is stopped just before the start of perfusion to minimize mixing.

6. The solution is bubbled gently with humidified 95% $O_2$/5% $CO_2$ to pH 7.4 (1–3 min) to attain steady state gas levels within the solution.
7. The test compound of interest is added to the perfusate at known concentration.
8. The solution is drawn up within a heated (37°C) glass syringe and maintained at 37°C throughout the experiment through the use of a circulating water bath and copper syringe jackets (*see* **Note 7**).
9. The perfusion syringe is placed within the perfusion pump and the pump is set for an infusion rate of 5–20 mL/min (*see* **Note 8**).
10. The perfusion catheter is attached to the perfusion syringe through a 20-gage needle and a three-way stopcock. The three-way stopcock can be used to attach a pressure transducer in line to monitor perfusion pressure during perfusion ***(2)***. In addition, a four-way valve can be put in line with a dual syringe infusion pump if "pre-perfusion" or "post-perfusion" with a different fluid is to be used as part of the uptake measurement.

### 3.4. Perfusion Procedure

1. The thoracic cavity is opened and the cardiac ventricles severed to stop blood flow from the systemic circulation.
2. Then, perfusion fluid is infused up the common carotid artery at a rate (5–20 mL/min) sufficient to perfuse the ipsilateral cerebral hemispheres at a normal pressure (approx 80–120 mm Hg) and the time of perfusion is monitored with a stopwatch. **Figure 2** illustrates a typical set up for a perfusion experiment.

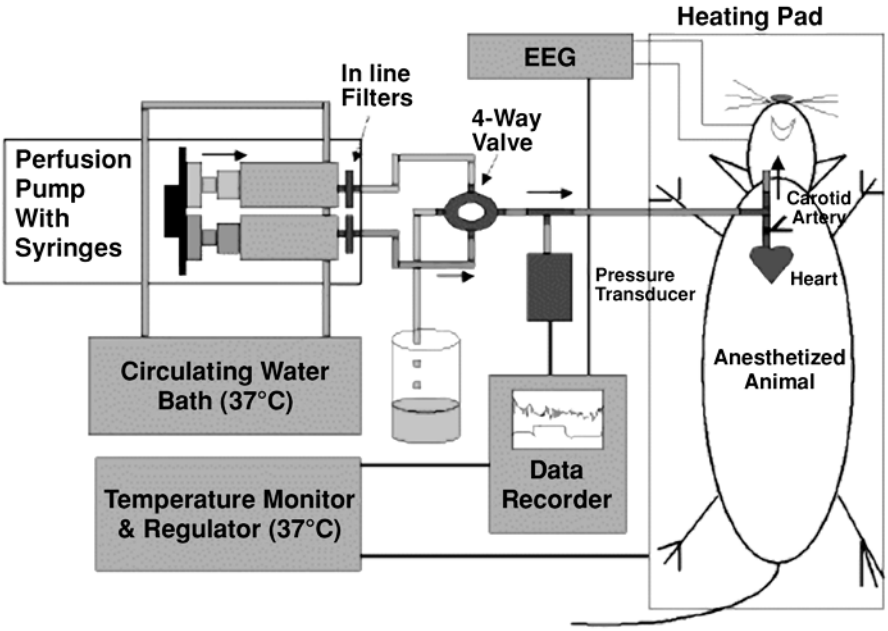


Fig. 2. Diagram illustrating the arrangement of the animal and the perfusion pump, temperature pad and regulator, and circulating water bath. Perfusion pressure can be measured with an in line pressure transducer linked to a recorder (Gould). The electroencephalograph can also be monitored with the same recorder used for the perfusion pressure.

3. Perfusion fluid that has passed through the brain vasculature bed and exited to the venous system accumulates in the thoracic cavity.
4. A prewash of the brain vasculature with drug or test-solute free fluid may be performed via perfusion for 10–45 s to clear the cerebral vessels of red cells and plasma constituents before onset with fluid containing test solute.
5. Because only one carotid artery is perfused, flow in the ipsilateral cerebral hemisphere exceeds that in the contralateral cerebral hemisphere by five- to tenfold. Bilateral perfusion can be achieved by cannulating and infusing fluid into both common carotid arteries (right and left).
6. Because the heart is stopped, there is no contribution to flow from the systemic circulation and the brain flow rate can be set at will for studies of the flow-dependent transport or metabolism.
7. At the end of the perfusion period, the pump is turned off and the animal is decapitated.
8. The brain is removed from the skull and placed on a filter paper (Whatman no. 1) moistened with 0.9% NaCl and sitting on a Petri dish cooled on ice (within an ice bucket).
9. Surface blood vessels are removed from the brain using cotton swabs.
10. The ipsilateral hemisphere is dissected into regions using surgical scalpel blades. Tissue sections are placed in sample tared vials and reweighed to obtain tissue weight.
11. Corresponding samples of perfusion fluid are also collected, placed in vials, and analyzed to obtain sample weight.
12. The total concentration of solute of interest in brain is then determined by scintillation counting, LC-MS, HPLC, GC or other suitable analytical means. When metabolism is a factor, it is often necessary to perform chromatography to carefully measure intact compound. Passage of solute across the blood-brain barrier can be demonstrated by separating

the cerebral microvessels from brain parenchyma using the capillary depletion technique ***(11,12)***.

13. At the end of perfusion, the amount of compound trapped in perfusion fluid in the brain blood vessels ($Q_{vas}$) is either removed from the brain by a short washout (10–15 s) with perfusion fluid that does not contain the compound of interest or it is measured indirectly using a vascular marker, such as [$^3$H]-sucrose or [$^{14}$C]-inulin which is included in the perfusion fluid ***(2,8)***.

### 3.5. Calculations

1. The BBB transfer coefficient for unidirectional uptake of solute into brain from perfusion fluid ($K_{in}$) is calculated as,

$$K_{in} = (Q_{tot} - V_v Q_{pf}) / C_{pf} T \quad (1)$$

where $C_{pf}$ = the concentration of solute in the perfusion fluid (mass drug/volume fluid), $Q_{tot}$ = total amount of solute in brain (mass/weight brain), $V_v$ = brain vascular volume (mL/g), and T= perfusion time.

2. The perfusion uptake time is calculated as the total time the solute was infused into the carotid artery minus the delay time it takes the perfusion fluid to reach the brain once the perfusion pump is started ***(2)*** (*see* **Note 9**).
3. The total quantity of solute in brain ($Q_{tot}$) is the sum of what has left the perfusate and crossed into brain ($Q_{br}$) and what remains in perfusate in the brain blood vessels ($Q_{vas}$) ***(2,5)*** (*see* **Notes 10** and **11**),

$$Q_{tot} = Q_{br} + Q_{vas} \quad (2)$$

4. $K_{in}$ can be converted into a cerebrovascular permeability-surface area product (PA) if the flow (F) of perfusion fluid in brain is known (*see* **Note 12**). PA is calculated from $K_{in}$ using the Crone Renkin equation as ***(2,3,5)***:

$$PA = -F \ln (1 - K_{in} / F) \quad (3)$$

5. F can be measured in brain perfusion experiments with radioactive diazepam or iodoantipyrine ***(2,3)***. [$^3$H]- or [$^{14}$C]Diazepam are most commonly used for such measurements and have the advantage that uptake is linear for 60 s ***(2,3,6)***. However, uptake with diazepam can become flow limited at high flow rates in the presence of plasma protein, as diazepam shows significant plasma protein binding.
6. The rate of drug uptake into brain ($J_{in}$) is given by the following equation,

$$J_{in} = K_{in} C_{pf} \quad (4)$$

7. If the interval over which brain uptake is unidirectional is unknown, this can be checked by measuring uptake over several different time periods and then plotting the data as $Q_{tot}/C_{pf}$ vs T and looking for the initial, linear portion of uptake curve ***(8,10–12)***. $K_{in}$ can be obtained by fitting the following equation to the linear portion of the uptake curve using least-squares regression:

$$Q_{tot} / C_{pf} = K_{in} T + Q_{vas} / C_{pf} \quad (5)$$

8. The brain perfusion technique can also be used to measure the rate of drug removal from brain by preloading the drug or solute of interest into the brain by vascular perfusion and then switching the perfusion fluid to drug-free fluid ***(10)***.

### 3.6. Applications

1. **Figure 3** illustrates the application of the *in situ* rat brain perfusion method to measure the rates of brain uptake and vascular intercepts of L-[$^{14}$C]alanine and [$^{14}$C]thiourea.

2. Data in **Fig. 3** are plotted as $Q_{br}/C_{pf}$ values to obtain effective "brain uptake spaces" with units of mL/g. Measured brain L-[$^{14}$C]alanine and [$^{14}$C]thiourea contents were corrected for residual intravascular radioactivity (i.e., tracer sitting within the brain blood vessels) using [$^{3}$H]inulin (1–2 μCi/mL) which was included in the perfusion fluid along with either [$^{14}$C]thiourea or L-[$^{14}$C]alanine (0.25–0.50 μCi/mL). The mean brain [$^{3}$H]inulin space equaled 0.0126 ± 0.0005 ml/g in the [$^{14}$C]thiourea and L-[$^{14}$C]alanine experiments and was comparable to previous reports ***(2)***. [$^{3}$H]Inulin has a very low BBB $K_{in}$ and is frequently used to measure brain vascular volume because it essentially does not cross the BBB in short experiments.
3. Brain uptake data were analyzed by linear regression to obtain slope and y-intercept values for the initial "linear" portion of the brain uptake graphs. Over the initial "linear" portion of the uptake plots, the slope is equivalent to the $K_{in}$ and the y-intercept is equivalent to the residual vascular space ($Q_{vas}/C_{pf}$), separate from that measured by [$^{3}$H]inulin.
4. The brain uptake of L-[$^{14}$C]alanine showed evidence of significant brain vascular accumulation (**Fig. 3A**), as noted by a positive y-intercept ($Q_{vas}/C_{pf}$). Rat brain was perfused with saline containing L-[$^{14}$C]alanine at tracer concentration in the absence of added unlabeled L-alanine or other amino acid competitors. Under these conditions, brain L-[$^{14}$C]alanine uptake slightly exceeded that of [$^{14}$C]thiourea, reaching brain/perfusion $Q_{br}/C_{pf}$ ratio values of approx 0.03 mL/g with a 60 s perfusion exposure. Brain L-[$^{14}$C]alanine uptake was linear and showed a best-fit regression $K_{in}$ of $3.9 \pm 0.3 \times 10^{-4}$ mL/s/g which decreased by 67% to $1.3 \pm 0.3 \times 10^{-4}$ mL/s/g upon addition of 10-m*M* unlabeled L-alanine to the perfusion fluid ($p < 0.05$). Such concentration-dependent reduction in $K_{in}$ is suggestive of brain L-[$^{14}$C]alanine uptake by a saturable transport carrier. However, unlike [$^{14}$C] thiourea, the calculated y-intercept $Q_{vas}/C_{pf}$ value for L-[$^{14}$C]alanine (0.0082 ± 0.0012 mL/g), significantly exceeded zero, suggesting rapid and selective accumulation of L-[$^{14}$C]alanine in the brain endothelium. This putative brain vascular component was concentration-dependent, as addition of 10-m*M* unlabeled L-alanine to the perfusate reduced the L-[$^{14}$C]alanine y-intercept to essentially zero (0.0006 ± 0.0016 mL/g) (*see* **Fig. 3A**).
5. [$^{14}$C]Thiourea, as a fairly polar nonelectrolyte, crossed the BBB slowly reaching a brain-perfusion fluid $Q_{br}/C_{pf}$ ratio of only 0.02 mL/g in 60 s of perfusion (**Fig. 3B**). The calculated BBB $K_{in}$ value obtained by least squares linear regression equaled $2.9 \pm 0.3 \times 10^{-4}$ mL/s/g. The calculated y-axis intercept value of $Q_{vas}/C_{pf}$ using **Eq. 5** equaled 0.0016 ± 0.0011 mL/g and did not differ significantly from zero ($p > 0.05$), suggesting that for [$^{14}$C]thiourea there is no additional binding or association with the vascular endothelium above that measured by [$^{3}$H]inulin. Thus, for [$^{14}$C]thiourea, [$^{3}$H]inulin appears to serve as a suitable marker of the vascular intercept space ($Q_{vas}/C_{pf} - Vv \approx 0$).
6. Given that the measured cerebral perfusion fluid flow rate (F = 0.1 mL/s/g) in the experiment greatly exceeded $K_{in}$ for both [$^{14}$C]thiourea and L-[$^{14}$C]alanine, no correction is required for flow-limited uptake and $K_{in}$ essentially equals PA for both tracers (*see* **Eq. 3**). In general, flow limitations only become important when $K_{in}/F \geq 0.2$ (*see* **Note 12**).
7. The results demonstrate the use of the *in situ* brain perfusion technique to quantitate and characterize solute transport across the BBB. This same method can be used in genetically modified rats and mice to study the effects of specific gene modifications on BBB function, such as knockout of the P-glycoprotein drug efflux transporter.

## 4. Notes

1. All surgery is performed using clean surgical technique. Deep anesthesia is required for surgery on the carotid circulation. Care should be taken to maintain anesthesia throughout the experiment and to insure that air bubbles and particulate matter is not introduced into

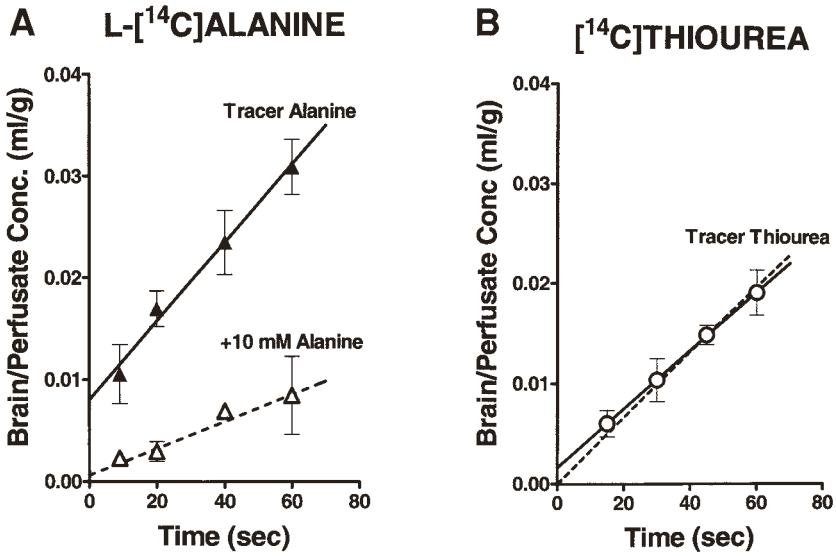


Fig. 3. Plot of the time course of brain uptake of L-[$^{14}$C]alanine (**left**) and [$^{14}$C]thiourea (**right**) during perfusion with physiologic saline fluid. The Y axis is the ratio of $Q_{br}$ divided by $C_{pf}$ to obtain the brain uptake "space" or brain/perfusate ratio (mL/g). The slope of the linear regression line is $K_{in}$, which is reduced for L-[$^{14}$C]alanine during perfusion with fluid containing 10-m*M* unlabeled L-alanine due to saturation of a transport carrier. The perfusate in both experiments contained [$^{3}$H]inulin to correct for intravascular tracer that is trapped in perfusion fluid within brain capillaries. $Q_{br}$ for L-[$^{14}$C]alanine and [$^{14}$C]thiourea was cal-culated as ($Q_{tot} - V_v Q_{pf}$) where $V_v$ was obtained from the $Q_{tot}/C_{pf}$ of [$^{3}$H]inulin. Data are mean ± SD for $n$ = 3–4.

the blood leading to the brain. The saline in the perfusion catheter should contain 100 IU/mL heparin sodium to minimize the formation of blood clots.

2. If only one of the two carotid arteries is ligated and cannulated, adequate blood flow to brain can be maintained prior to perfusion in normal animals via cross over of blood from the contralateral carotid artery at the Circle of Willis, which joins the vessels that feed the brain ***(2,3)***. This prevents hypoxic brain damage in the intervening minutes as the perfusion cannula is connected to the pump and the perfusion is begun.
3. Some groups have found that it possible to perfuse via the left cardiac ventricle or aorta ***(13–15)***. Such systems provide equal perfusion to both hemispheres of the brain, but require higher infusion rates.
4. A number of perfusion fluids have been developed for use with the *in situ* brain perfusion technique. The optimum fluid depends upon the experimental goal. The BBB will accept considerable variation in perfusate composition for short intervals (<5 min) as long as the pH (7.0–7.5) and osmolarity (280–320 mOsm/kg) are maintained within normal limits.
5. Serum proteins are not required in the perfusate for oncotic pressure, as the brain capillaries have a low permeability to both large and small solutes. However, serum proteins often serve critical roles with regard to drug binding and transport, and may be added for that purpose, as well as to maintain oncotic balance in the "nonbarrier" regions or tissues. Albumin is frequently used for this purpose (2.7–4%). High molecular weight, nonprotein polymers, including dextran, can also be used.
6. For long perfusions (>60 s), an oxygen-carrying agent should be added to the perfusate to sustain normal cerebral metabolic and electrical activity. Most use washed erythrocytes to

a hematocrit of 20–40% or whole rat blood. With good oxygenation, cerebral oxygen metabolism and EEG can be maintained for more than 1 h.

7. In-line filters are valuable for removal of particulate matter and clots from the perfusion fluid ***(2)***. But, care should be taken because they also can bind drugs (e.g., diazepam) and lead to lower concentration of test drug in the infusion solution coming out of the filter.
8. The recommended infusion rates for the modified Takasato method (cannulated common carotid artery—ligated external carotid artery) with saline perfusion fluid is 5–20 mL/min. In general, saline has a lower viscosity than blood and requires a higher infusion rate to obtain the same pressure. Pressures in excess of 200 mm Hg should be avoided because they may damage the BBB.
9. Often 1–5 s are required after start of the perfusion pump for the perfusion fluid to reach the brain. This delay time should be subtracted from the total perfusion time to obtain the net perfusion time (i.e., the time that perfusion fluid was actually passing through the brain). The delay time will differ depending on the size and length of the perfusion catheter and the infusion rate. The delay time can be measured by creating a cranial window in the skull of an animal immediately before perfusion and then determining how long it takes for the blood to clear from the cerebrovascular surface vessels after start of the perfusion pump.
10. Investigators should conduct all perfusion experiments so that more than half of the drug in brain has crossed the BBB ($Q_{tot}/C_{pf} \geq 2 \times Q_{vas}/C_{pf}$). Otherwise, small errors in intravascular drug distribution can appreciably affect the brain permeability calculations ***(3,5)***. Further, it may be appropriate in some instances to confirm that the test solute does not modify BBB permeability. The latter has been shown to occur for some vasoactive peptides and drugs.
11. Some solutes bind significantly to the luminal surface of the cerebral blood vessels and can lead to overestimation of brain uptake if an appropriate vascular correction is not made. Drugs that exhibit this behavior tend to be either cationic (e.g., choline) ***(10,16)*** or highly hydrophobic. The loosely bound vascular drug can be removed by a brief wash (10–15 s) with drug-free saline ***(10)***. Bovine serum albumin (2.7 g/dL) can be added to the perfusion wash solution to enhance the efficiency of vascular drug removal.
12. If the drug or test solute binds minimally to plasma proteins ($f \approx 1.0$) and the transfer constant ($K_{in}$) is small relative to the flow rate (F)(i.e., $K_{in} < 0.2$ F), then $K_{in}$ essentially equals the PA product (within <10% error) and is essentially flow independent ***(5)***. On the other hand, if the $K_{in}$ is large relative to flow ($K_{in} \geq 0.2$ F), then uptake will depend both on flow and permeability and flow measurements will be required in order to obtain accurate estimates of BBB PA.

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